# Testing

Two suites for two different jobs, and a third thing that is not a suite at all.

| Command | What runs | Needs | Takes |
|---|---|---|---|
| `npm test` | The rules, as plain functions | nothing | ~2 seconds |
| `npm run test:integration` | The API against a real Postgres | Docker | ~30 seconds |
| `npm run test:all` | Both | Docker | ~35 seconds |
| `npm run test:db:stop` | Removes the test database container | Docker | — |
| `npm run test:preview` | A walk through the customer app on the demo shop | the preview running | ~2 seconds |

**950 tests today: 351 on the rules, 599 on the API.** Both suites must be green before anything is pushed; the preview walk is run by hand when the demo shop is up.

Every one of them also runs on every push, in GitHub Actions: types, lint and the rules in one job, the whole API against a real Postgres in another, and the production build against an empty database — which proves the schema still applies to a fresh shop.

## The rules ( `src/**/*.test.ts` )

Pure functions with no database: free slots and opening hours, how a visit was paid, session lengths, phone numbers, receipts, who may use which door — and since, the loyalty card's arithmetic, what a rating averages to, who belongs in which customer group, when a birthday falls in shop time, what the insight screens count, what a customer always books, how much of a deposit to ask for, who a freed chair suits, and when a miss stops counting. Twenty-five files, fast enough to keep running while you work ( `npm run test:watch` ).

These exist because the awkward parts of this system are arithmetic — a buffer around an appointment, a mixed payment that must add up, a clock that changes twice a year. Arithmetic is cheap to test and expensive to get wrong.

## The API (`tests/*.test.ts`)

The route handlers, against a real Postgres in a container, with the session mocked to whichever role the test is about. Thirty-three files, of which the ones worth knowing about are:

- `booking-api.test.ts` — booking, clashes, opening hours, buffers, time off, moving, cancelling, and who may use the shop's diary.
- `customer-api.test.ts` — signing up, what a customer may book, the notice period, the limit on waiting bookings, cancelling their own and not others', and the wall between customer and shop.
- `money-api.test.ts` — a visit rung up at the till through the drawer, the day journal and the period report: cash, card, InstaPay, mixed, tips, expenses, advances.
- `auth.test.ts` — signing in by email or phone, and how long a login lasts.
- `checkout-api.test.ts`, `queue-api.test.ts`, `diary-summary-api.test.ts` — ringing a booking up, the live queue, and the day at a glance.
- `barber-app.test.ts`, `barber-visibility.test.ts` — a barber's own day and earnings, and who the customer app is allowed to show.
- `offers-api.test.ts`, `push-api.test.ts`, `reviews-api.test.ts` — offers, notifications and reminders, and reviews: who may leave one, one per visit, and the owner's power to hide but not delete.
- `loyalty-api.test.ts` — stamps counted from paid visits only, a card that cannot be cashed in twice, and the proof that giving a reward moves no money.
- `insights-api.test.ts` — the owner's read of the shop: the window, who may ask, and the proof that a long-standing customer is never counted as a new face.
- `drawer-api.test.ts` — opening, moving money by hand, and closing against what was counted. No second session over an open one, nothing added to a drawer that is shut.
- `payroll-api.test.ts` — penalties, advances, expenses and the salary sheet they land on. A cashier may take an expense; only the owner may fine a barber.
- `shop-setup-api.test.ts` — the price list, the customer book, and the PINs in front of a discount and an edit.
- `warehouse-api.test.ts` — a purchase raises the balance, a transfer moves it without creating any, nothing can be issued that is not there.
- `reporting-api.test.ts` — the overview, discounts, absences, and the activity trail, which must outlive the thing it recorded and stay unreadable to the people it watches.
- `retail-api.test.ts` — selling a product: the money and the shelf moving together, and never selling what is not there.
- `waitlist-api.test.ts` — a cancellation offered to the longest waiter it suits, one offer at a time.
- `deposits-api.test.ts` — the one that matters most for money: a deposit is never revenue when it arrives, and is counted once rather than twice.

- `time-off-api.test.ts` — asking blocks nothing until somebody answers.
- `wall-api.test.ts`, `portfolio-api.test.ts` — what the screen on the wall may say about a customer, and whose photos are whose.
- `cut-reports-api.test.ts` — who may write one (a barber about their own chair only) and who may read the shop's (the owner alone), on a booked cut and on a walk-in alike, with exactly one of the two named.
- `cuts-wanted-api.test.ts` — the photos are the customer's in both directions, and they reach the barber's day screen.
- `walk-in-reviews-api.test.ts` — a cut rated whichever way it arrived, once and only once, and never offered twice when a booking was rung up.
- `walk-in-usual-api.test.ts` — a habit read from walk-ins as well as bookings, counted once when a booking was rung up, and a live queue never invented out of an empty one.
- `customer-history-api.test.ts` — a customer sees every one of their own haircuts and none of anybody else's, a page at a time, and never the tip, the discount or how it was paid.
- `customer-roster-api.test.ts` — who is in for the week, why nobody is told the reason, and the deposit address going only to somebody actually asked for one.
- `barber-profile-api.test.ts` — a barber's face: saved, cleared, refused when it is not a picture or is too big to ride along with the catalogue, and the bio a new barber used to arrive without.
- `barber-hours-api.test.ts` — a barber's own week: it narrows the shop's day and never widens it, a whole week or none, and the slot maths and roster both obey it.
- `barber-login-api.test.ts` — a login has one way in, a PIN is six digits and stored hashed, and the phone that was typed signs the barber in however they wrote it.
- `segments-api.test.ts` — the groups, and the guardrails on writing to one: who may send, once every twelve hours, and only to people who asked for notifications.
- `referrals-api.test.ts` — a code that survives being written down badly, a referral earned only by a friend who actually came in, a reward given once, and a birthday greeting that goes out once a year.

A mock cannot tell you that two people took the same slot, or that the drawer and the report disagree. These can.

The database is a throwaway container ( `escape-test-db` , port 55499), started on the first run and reused afterwards. Every table is emptied between tests, and the connection string is built in `tests/db.ts` — the shop's real database is never reachable from a test.

## The walk through the preview ( `tests/manual` )

`npm run test:preview` walks the customer app end to end against the demo shop on your own machine: the profile and its full loyalty card, the price list, free times on a real day, a booking that it

then cancels, a review, the referral code and the notification key. It prints what it found rather than only passing.

It is kept out of the normal suite on purpose. The preview is not always up, and a green suite must mean the code is sound rather than that somebody happened to have Docker running.

## Writing another one

- A new rule starts as a function in `src/lib` with a unit test, before any screen.
- A new endpoint gets an integration test covering what it does **and** who is refused. Every route was written with both, and one of those refusal tests is what caught a customer being able to open the shop's own appointments screen.
- Reach for real rows over mocks. The bugs live in what the database actually did.

## What is not covered yet

- The screens themselves: no browser tests. Checked by hand for now.
- The Excel and PDF exports beyond the receipt text.
- The warehouse and payroll screens, which predate this suite and are covered only through the money tests they feed.